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$$Q. \quad T \cdot P^3 - 7P^5 \quad \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

$$(i) \quad T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} 0x_1 + x_2 + x_3 \\ x_1 + 0x_2 - x_3 \\ -x_1 - x_2 + 0x_3 \end{bmatrix}$$

$$\begin{bmatrix} x_2 + x_3 \\ x_1 - x_3 \\ -x_1 - x_2 \end{bmatrix}$$

$$T(x_1, x_2, x_3) = (x_2 + x_3, x_1 - x_3, -x_1 - x_3)$$

(ii) Given $v_1 = (0, 1, 2)$, $v_2 = (1, 1, 1)$, $v_3 = (1, 0, -2)$

$$T(v_1) = T(0, 1, 2) = (3, -2, -1)$$

$$T(v_2) = T(1, 1, 1) = (2, 0, -2)$$

$$T(v_3) = T(1, 0, -2) = (-2, 3, -1)$$

Matrix of A of T becomes

$$A = \begin{bmatrix} 3 & 2 & -2 \\ -2 & 0 & 3 \\ -1 & -2 & -1 \end{bmatrix}$$